Resolving the Remaining Cases of Carlitz-Extensiveness over Finite Fields using SageMath

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Main goal of this talk

The question

For which finite field extensions $\mathbb{F}_{q^n}/\mathbb{F}_q$ does there exist, for *every* $z \in \mathbb{F}_{q^n}$, an element $x \in \mathbb{F}_{q^n}$ that is both primitive *and* a γ_z-generator?

- Open since Hsu & Nan (2011); narrowed by Hachenberger (2016).
- Six pairs (q, n) remained unresolved.

What we show

Using SageMath, we settle the six remaining cases. Combined with earlier results, this yields the complete classification:

$$\mathbb{F}_{q^n}/\mathbb{F}_q \text{ is Carlitz-extensive} \iff (q, n) \neq (2, 2).$$

Lenstra–Schoof	Hsu & Nan	Hachenberger	Gruber	this work
1987	2011	2016	2016	2026

Overview

1. The Primitive Normal Basis Theorem
2. The Hsu–Nan Generalization (2011)
3. The Hachenberger Generalisation (2016)
4. Reduction Tools
5. Algorithm
6. Results

The Primitive Normal Basis Theorem

Two classical kinds of generators

Let $\mathbb{F}_{q^n}/\mathbb{F}_q$ be a finite field extension of degree n.

Primitive element ($\to$ multiplicative field structure)

An element $\alpha \in \mathbb{F}_{q^n}$ is called **primitive** if it generates the multiplicative group:

$$\mathbb{F}_{q^n}^{\times} = \langle \alpha \rangle = \{\alpha, \alpha^2, \ldots, \alpha^{q^n-1}\}.$$

Equivalently, $\mathrm{ord}(\alpha) = q^n - 1$.

Normal element ($\to$ additive vector space structure)

An element $\alpha \in \mathbb{F}_{q^n}$ is called **normal** (over $\mathbb{F}_q$) if the iterates of the Frobenius automorphism form an $\mathbb{F}_q$-basis of $\mathbb{F}_{q^n}$:

$$\{\alpha,\ \sigma(\alpha),\ \sigma^2(\alpha), \ldots, \sigma^{n-1}(\alpha)\} = \{\alpha,\ \alpha^q,\ \alpha^{q^2}, \ldots, \alpha^{q^{n-1}}\} \quad \text{is an } \mathbb{F}_q\text{-basis of } \mathbb{F}_{q^n}.$$

The Primitive Normal Basis Theorem

Theorem (Lenstra & Schoof, 1987)

For every finite field extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ there exists an element $\alpha \in \mathbb{F}_{q^n}$ which is simultaneously primitive and normal over $\mathbb{F}_q$.

- Such an α generates $\mathbb{F}_{q^n}^\times$ multiplicatively and $\mathbb{F}_{q^n}$ additively (via its Frobenius basis).
- Original proof combines character sums with a finite computer search.
- A computer-free proof was given by Cohen & Huczynska (2003).

H. W. Lenstra Jr. and R. J. Schoof, *Primitive normal bases for finite fields*, Math. Comp. **48** (1987), 217–231.

S. D. Cohen and S. Huczynska, *The primitive normal basis theorem—without a computer*, J. London Math. Soc. (2) **67** (2003), 41–56.

The Hsu–Nan Generalization (2011)

Hsu & Nan: a Carlitz-module viewpoint

- Hsu and Nan (2011) reformulated Primitive Normal Basis Theorem in language of finite Carlitz modules
- **Main idea**: Can the *Frobenius automorphism* $\sigma : \mathbb{F}_{q^n} \to \mathbb{F}_{q^n}$, $x \mapsto x^q$ be replaced by a more general $\mathbb{F}_q$-linear map?
- **Generalisation**:

$$\text{Frobenius automorphism} \longrightarrow \gamma_z\text{-endormorphism}$$

$$\text{Primitive normal elements} \longrightarrow \text{Primitive } \gamma_z\text{-generators}$$

The endomorphism γ_z

Definition

For $z \in \mathbb{F}_{q^n}$, define the map

$$\gamma_z : \mathbb{F}_{q^n} \longrightarrow \mathbb{F}_{q^n}, \qquad \gamma_z(x) \;=\; z \cdot x \;+\; x^q.$$

- γ_z is an $\mathbb{F}_q$-linear endomorphism of $\mathbb{F}_{q^n}$
- Special case $z = 0$:
  $\gamma_0(x) = x^q = \sigma(x)$ is the Frobenius automorphism. Hence, γ_z generalises the Frobenius automorphism.
- γ_z is not always bijective:
  e.g. $\gamma_{-1}(1) = (-1) \cdot 1 + 1^q = -1 + 1 = 0$, so $1 \in \ker \gamma_{-1}$.

The minimal polynomial of γ_z

Definition

The **minimal polynomial** $\mathrm{m}_{\gamma_z} \in \mathbb{F}_q[x]$ of γ_z is the unique monic polynomial of smallest degree satisfying

$$\mathrm{m}_{\gamma_z}(\gamma_z) = 0 \quad \text{in } \mathrm{End}_{\mathbb{F}_q}(\mathbb{F}_{q^n}).$$

Lemma (Hsu–Nan)

Let $z \in \mathbb{F}_{q^n}$ have minimal polynomial $f(x) \in \mathbb{F}_q[x]$ of degree $d \mid n$. Then

$$\mathrm{m}_{\gamma_z}(x) = f(x)^{n/d} - 1.$$

Example: For $z = 0$: $f(x) = x$ and $d = 1$, so $\mathrm{m}_{\gamma_0}(x) = x^n - 1$. This is indeed the minimal polynomial of the Frobenius $\sigma = \gamma_0$.

γ_z-generators

Definition (γ_z-generator)

An element $\alpha \in \mathbb{F}_{q^n}$ is a γ_z-**generator** if

$$\{\alpha,\ \gamma_z(\alpha),\ \gamma_z^2(\alpha), \dots, \gamma_z^{n-1}(\alpha)\} \quad \text{is an } \mathbb{F}_q\text{-basis of } \mathbb{F}_{q^n}.$$

Two equivalent characterisations

- α generates $\mathbb{F}_{q^n}$ as an $\mathbb{F}_q[x]$-module via γ_z, i.e., $\mathbb{F}_q[x] \cdot \alpha = \mathbb{F}_{q^n}$ with $f(x) \cdot \alpha = f(\gamma_z)(\alpha)$
- The monic, minimal degree polynomial $\mathrm{ord}_{\gamma_z}(\alpha)$ s.t. $\mathrm{ord}_{\gamma_z}(\alpha)(\gamma_z)(\alpha) = 0$ is equal to m_{γ_z}, i.e. $\mathrm{ord}_{\gamma_z}(\alpha) = \mathrm{m}_{\gamma_z}$

- For every $z \in \mathbb{F}_{q^n}$, a γ_z-generator always exists (analogue of the normal basis theorem).
- However: γ_z-generators are often not primitive $\rightarrow$ should be simultaneously primitive and γ_z-generator

Definition (Carlitz-extensiveness)

The extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is **Carlitz-extensive** if for every $z \in \mathbb{F}_{q^n}$, there exists a primitive γ_z-generator $\alpha \in \mathbb{F}_{q^n}$, i.e. α is primitive in $\mathbb{F}_{q^n}$ and a γ_z-generator.

Theorem (Lenstra & Schoof, 1987)

For every finite field extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ there exists an element $\alpha \in \mathbb{F}_{q^n}$ which is simultaneously primitive and normal over $\mathbb{F}_q$.

Bridge to the classical theorem

Specialisation $z = 0$

$$\gamma_0(\alpha) = \sigma(\alpha) = \alpha^q,$$

$$\{\alpha, \gamma_0(\alpha), \ldots, \gamma_0^{n-1}(\alpha)\} = \{\alpha, \alpha^q, \ldots, \alpha^{q^{n-1}}\}.$$

Hence,

$$\boxed{\alpha \text{ is a primitive } \gamma_0\text{-generator} \iff \alpha \text{ is primitive and normal.}}$$

Consequence: If every extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Carlitz-extensive, the Primitive Normal Basis Theorem follows as a corollary. Carlitz-extensiveness is therefore a strict generalisation.

Hsu & Nan: open cases

Hsu & Nan reduced Carlitz-extensiveness problem to finitely many, namely 63, pairs (q, n) which their character-sum estimates could not handle (too small for asymptotic estimates):

q	n_q
2	2–12, 14, 15, 16, 18, 20, 21, 22, 24, 28, 30, 36
3	2, 3, 4, 6, 8, 9, 10, 14, 15, 16, 18, 20, 22, 24, 30
4	3–10, 12, 15
5	2, 4, 5, 6, 8, 10, 12
7	3, 6
8	4, 8
9	2, 4, 6, 8
11	6
13	4

Table 3 from Hsu & Nan, J. Number Theory 131 (2011), 146–157.

Counterexample: $\mathbb{F}_4/\mathbb{F}_2$ is not Carlitz-extensive

Setup. $\mathbb{F}_{q^n} = \mathbb{F}_{2^2} = \mathbb{F}_4 = \mathbb{F}_2[x]/(x^2+x+1)$, $\mathbb{F}_q = \mathbb{F}_2$. α root of x^2+x+1, so $\alpha^2 = \alpha+1$.

- $|\mathbb{F}_4^\times| = 3$ is prime $\implies$ primitive elements are $\{\alpha, \alpha+1\}$
- x is a γ_z-generator $\iff$ $\{x, \gamma_z(x)\}$ is an $\mathbb{F}_2$-basis, where $\gamma_z(x) = z\cdot x + x^2$

z	$\{\alpha, \gamma_z(\alpha)\}$	$\{\alpha+1, \gamma_z(\alpha+1)\}$
0	$\{\alpha, \alpha+1\}$ ✓	$\{\alpha+1, \alpha\}$ ✓
1	$\{\alpha, 1\}$ ✓	$\{\alpha+1, 1\}$ ✓
α	$\{\alpha, 0\}$ ✗	$\{\alpha+1, \alpha+1\}$ ✗
$\alpha+1$	$\{\alpha, \alpha\}$ ✗	$\{\alpha+1, 0\}$ ✗

Conclusion: For $z \in \{\alpha, \alpha+1\}$, only 1 but no primitive element is a γ_z-generator

The Hachenberger Generalisation (2016)

Hachenberger: cyclic vector spaces

Hachenberger (2016) places the problem in a wider framework.

- Replace γ_z by any $\mathbb{F}_q$-vector space endomorphism τ of $\mathbb{F}_{q^n}$ for which a τ-generator exists
- A **primitive** τ-**generator** generates $\mathbb{F}_{q^n}^\times$ multiplicatively and $\mathbb{F}_{q^n}$ as an $\mathbb{F}_q[x]$-module under τ

Specialisations:

- $\tau = \sigma \;\Rightarrow\;$ primitive normal element $\to$ Lenstra–Schoof
- $\tau = \gamma_z,\ z \in \mathbb{F}_{q^n} \;\Rightarrow\;$ primitive γ_z-generator $\to$ Hsu–Nan

Progress:

- 41 of the open Hsu–Nan cases resolved
- Master's thesis of Thomas Gruber reduced the list further by 16 pairs

Hachenberger, Australasian J. Combin. 64 (2016), 289–326.

The remaining 6 cases

After the work of Hachenberger and Gruber, only six pairs (q, n) remained unresolved:

q	n
2	18, 20, 24
4	10, 12
8	8

- Too large for naive brute force: q^n up to $2^{24} \approx 1.7 \cdot 10^7$.
- Too small for asymptotic or structural arguments

Goal of this work: Settle each of these six cases by an explicit, machine-verifiable computation in SageMath

Reduction Tools

Strategy at a glance

The cases range from $q^n = 2^{18} \approx 2.6 \cdot 10^5$ up to $2^{24} \approx 1.7 \cdot 10^7$. A brute-force search over all pairs $(z, x) \in \mathbb{F}_{q^n}^2$ is infeasible. We reduce the search drastically using:

1. Cyclotomic cosets: one z per coset suffices (outer loop), simplifies calculating of the minimal polynomial
2. Projective space: one x per projective line suffices (inner loop), facilitates finding a primitive γ_z-generator, once a γ_z-generator is found
3. Minimal-polynomial factorisation: fast algebraic test to check the γ_z-generator-property, more efficient than e.g. the basis check

Each tool decreases the number of computations needed, making it possible to run the algorithm in a realistic time span.

Cyclotomic cosets

Definition

A cyclotomic coset (modulo $q^n - 1$) is an orbit of multiplication by q on $\mathbb{Z}/(q^n - 1)\mathbb{Z}$:

$$C_k = \{k,\ kq,\ kq^2, \dots\} \pmod{q^n - 1},$$

where $0 \le k \le q^n - 2$.

Their sizes $|C_k|$ are divisors of n.

Let $g \in \mathbb{F}_{q^n}$ be a primitive element. Exponentiating turns each cyclotomic coset C_k into a Frobenius orbit in $\mathbb{F}_{q^n}^\times$:

$$\mathcal{O}(g^k) = \{g^j \mid j \in C_k\} = \{g^k, \sigma(g^k), \sigma^2(g^k), \dots\}.$$

Computing the minimal polynomial given the cyclotomic coset

Lemma (Minimal polynomial via orbits)

The minimal polynomial of g^k over $\mathbb{F}_q$ is

$$f_{g^k}(x) = \prod_{i=0}^{|C_k|-1} \left(x - \sigma^i\left(g^k\right)\right) = \prod_{j \in C_k} \left(x - g^j\right).$$

- Given the cyclotomic coset of k, resp. the Frobenius orbit of g^k, we can thus easily compute the minimal polynomial of g^k.

Frobenius compatibility of γ_z

Lemma (Conjugation lemma)

For every $z \in \mathbb{F}_{q^n}$ and every $i \in \mathbb{Z}$,

$$\gamma_{\sigma^i(z)} = \sigma^i \circ \gamma_z \circ \sigma^{n-i}.$$

Corollary

If α is a primitive γ_z-generator, then $\sigma^i(\alpha)$ is a primitive $\gamma_{\sigma^i(z)}$-generator.

- Primitivity is preserved as σ^i is a field automorphism (preserves multiplicative order).
- γ_z-generator property follows from the conjugation lemma:

$$\{\sigma^i(\alpha), \gamma_{\sigma^i(z)}(\sigma^i(\alpha)), \dots\} = \sigma^i\{\alpha, \gamma_z(\alpha), \dots\}.$$

Reduction: It suffices to test existence of a primitive γ_z-generator for one z per Frobenius orbit.

Projective space & primitive projective points

Set $N = q^n - 1$ and $Q = \dfrac{q^n - 1}{q - 1}$.

Projective space

The projective $(n-1)$-space over $\mathbb{F}_q$ is

$$\mathbb{P}^{n-1}_{\mathbb{F}_q} \;=\; \{[x] \mid x \in \mathbb{F}_{q^n} \setminus \{0\}\}, \qquad [x] = \mathbb{F}_q^\times \cdot x.$$

Equivalently, $\mathbb{P}^{n-1}_{\mathbb{F}_q}$ is the cyclic quotient $\mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times$ of order Q.

Definition (primitive projective point)

A point $[x] \in \mathbb{P}^{n-1}_{\mathbb{F}_q}$ is **primitive** if its class generates the cyclic group $\mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times$:

$$\{[x], [x]^2, \ldots, [x]^Q\} \;=\; \mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times.$$

From projective to affine primitives

Let $g \in \mathbb{F}_{q^n}$ be a (fixed) primitive element. Then:

Lemma

$[g^k] \in \mathbb{P}^{n-1}_{\mathbb{F}_q}$ is a primitive projective point $\iff$ $\gcd(k, Q) = 1$
$\iff$ There exists $\lambda \in \mathbb{F}_q^\times$ such that $\lambda g^k \in \mathbb{F}_{q^n}^\times$ is primitive.

Corollary

Every primitive projective point is represented by a primitive element of $\mathbb{F}_{q^n}^\times$ on its line. Conversely, every primitive element of $\mathbb{F}_{q^n}^\times$ projects to a primitive projective point.

Consequence: We can check only primitive projective points as they represent all primitive points.

Lines preserve the γ_z-generator-property

Lemma ($\mathbb{F}_q^\times$-invariance)

Let $z, x \in \mathbb{F}_{q^n}$ and $\lambda \in \mathbb{F}_q^\times$. Then

$$x \text{ is a } \gamma_z\text{-generator} \iff \lambda x \text{ is a } \gamma_z\text{-generator.}$$

Proof sketch:

γ_z is $\mathbb{F}_q$-linear, so $\gamma_z^i(\lambda x) = \lambda \gamma_z^i(x)$.

Thus, $\{\lambda x, \gamma_z(\lambda x), \ldots, \gamma_z^{n-1}(\lambda x)\}$ $\mathbb{F}_q$-basis $\iff$ $\{x, \gamma_z(x), \ldots, \gamma_z^{n-1}(x)\}$ $\mathbb{F}_q$-basis.

Projective space relation: Property of being γ_z-generator depends only on projective point $[x]$

Consequence: Having found a primitive projective point that is a γ_z-generator, we can be sure that there exists a primitive γ_z-generator in the same equivalence class.

Fast γ_z-generator-property test via the minimal polynomial

Let $\mathrm{m}_{\gamma_z}(x) \in \mathbb{F}_q[x]$ be the minimal polynomial of γ_z and write

$$\mathrm{m}_{\gamma_z}(x) = p_1(x)^{e_1}\, p_2(x)^{e_2} \cdots p_k(x)^{e_k}$$

with $p_1, \ldots, p_k$ distinct irreducible factors in $\mathbb{F}_q[x]$.

Generator test

x is not a γ_z-generator $\iff$ There exists $i \in \{1, \ldots, k\}$ with

$$\left(\frac{\mathrm{m}_{\gamma_z}}{p_i}\right)(\gamma_z)(x) = 0.$$

- Replaces basis-determinant computations by a small number of polynomial evaluations
- Horner's method used for a fast evaluation of the polynomial
- The $\mathbb{F}_q$-linear map γ_z is implemented as matrix using some fixed basis $\rightarrow$ vector-matrix multiplications for each $i \in \{1, \ldots, k\}$

Algorithm

The algorithm in one slide

Input. A pair (q, n). Output. `True` / `False` for Carlitz-extensiveness.

Preparation.

1. Fix a primitive element $g \in \mathbb{F}_{q^n}^{\times}$; set $N = q^n - 1$, $Q = N/(q-1)$.
2. Precompute representatives of the primitive projective points:

$$\mathcal{R} = \{\, g^k : 1 \le k < Q,\ \gcd(k, Q) = 1 \,\}.$$

3. Handle $z = 0$ separately (Frobenius case): search $\mathcal{R}$ for a primitive γ_0-generator.

Main loop. For each q-cyclotomic coset of $\mathbb{Z}/N\mathbb{Z}$:

1. Pick a representative exponent e; set $z = g^e$.
2. Compute f_z (minimal polynomial of z, via its cyclotomic coset) and factor $\mathrm{m}_{\gamma_z} = f_z^{n/\deg f_z} - 1 = \prod_i p_i^{e_i}$.
3. Build the matrix A of γ_z, then $H_i = (m_{\gamma_z}/p_i)(A)$ for each i.
4. Scan $\mathcal{R}$: if $g^k \cdot H_i \neq 0$ for *all* i, then $\alpha = \lambda g^k$ is a primitive γ_z-generator.
5. If $\mathcal{R}$ is exhausted with no hit: return `False`.

If every coset succeeds: return `True`.

Results

Main result

Theorem (Hellwig & Schaller, 2026)

Each of the six remaining pairs

$$(q, n) \in \{(2, 18), (2, 20), (2, 24), (4, 10), (4, 12), (8, 8)\}$$

is Carlitz-extensive.

Combined classification

A finite-field extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Carlitz-extensive if and only if $(q, n) \neq (2, 2)$.

Equivalently, $\mathbb{F}_4/\mathbb{F}_2$ is the unique non-Carlitz-extensive extension among all finite extensions of finite fields.

All computations were carried out with a Python 3 Kernel importing the sage module. The six cases together need around 18h to run (on one core).

Lenstra–Schoof	Hsu & Nan	Hachenberger	Gruber	**this work**
1987	2011	2016	2016	**2026**

Alternative code: generator-first search

Basic idea

Instead of first scanning primitive projective points, the alternative code first finds one γ_z-generator u and then searches among

$$a(\gamma_z)(u), \qquad a(x) \in \mathbb{F}_q[x] \text{ monic}, \quad \deg(a(x)) < n, \quad \gcd(a(x), \mathrm{m}_{\gamma_z}(x)) = 1.$$

For each candidate, it tests whether some $\lambda \cdot a(\gamma_z)(u)$, $\lambda \in \mathbb{F}_q^\times$, is primitive.

- Takes a bit longer, but verifies the results
- Exhaustive run for the six remaining cases $\approx$ 21h

Code and additional material

Repository

All codes, including pseudocodes, benchmarking files, and additional information, are available on GitHub:

`github.com/Smile4heWorld/Carlitz-extensiveness`

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Thank you!

Questions?